B.Sc. (General/Special) Degree Programme
Faculty of Applied Sciences
University of Sri Jayewardenepura

ICT 1022 2.0 Computer System Architecture

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Digital Computer

Digital Computer

- A digital computer is **a machine that can solve problems for people by carrying out instructions given to it**.
- A sequence of instructions describing how to perform a certain task is called a **program**.

What is Program?

- Computer program is the composition of sequences of instructions that computers can follow to perform task.

What is Software?

- Software contains the instructions that tell a computer—or a computerized device—what to do.

What is a programming language?

- Computers do not understand human languages, so programs must be written in a language a computer can use.

Digital Computer

- The **electronic circuits of each computer can recognize and directly execute** a limited set of simple instructions.
- All its programs must be converted into simple instructions set before they can be executed.
- Basic instructions are **more complicated** than
  - Add two numbers.
  - Check a number to see if it is zero.
  - Copy a piece of data from one part of the computer's memory to another.

What is Machine Language?

- Native language of computers.
- It is composed of digital binary numbers.

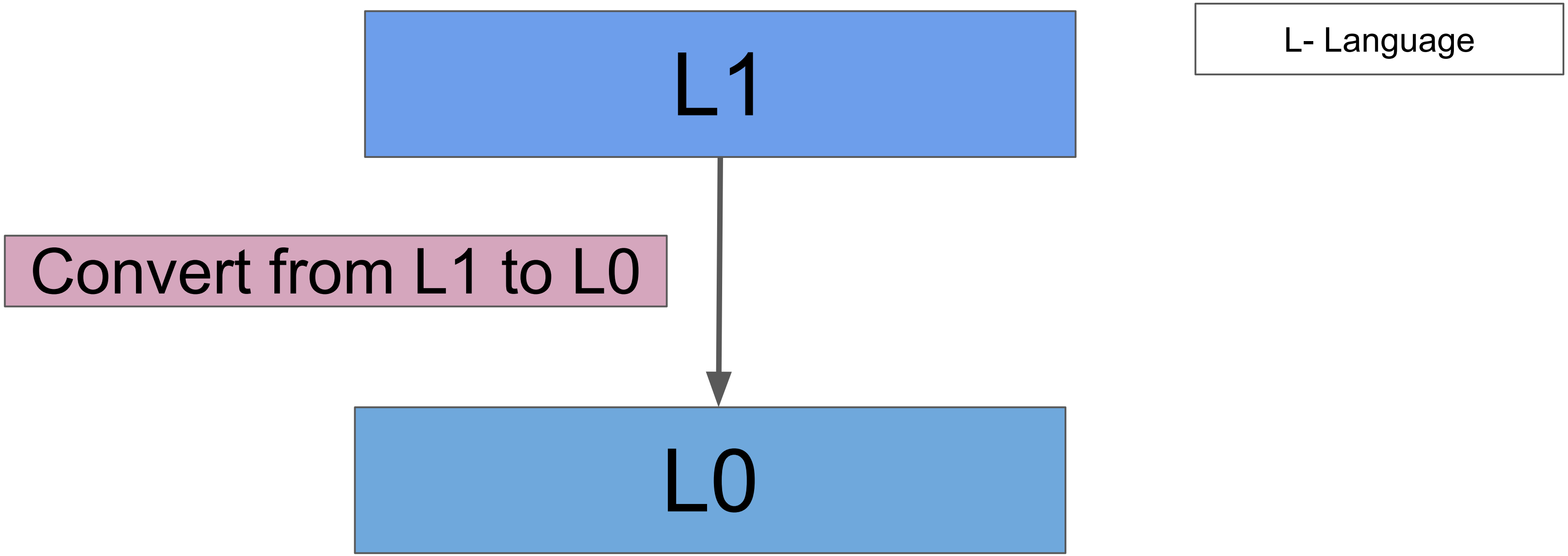
For example, to add two numbers, you might have to write an instruction in binary code.

1101101010011010

Machine language

- When designing a new computer, it must decide what instructions to include in its machine language.
- It considers
  - **simplicity,**
  - **consistent .**
- Aim to,
  - **reduce the complexity.**
  - **reduce the cost of the electronics needed.**
- However, machine language is difficult and tedious for people to use.

Designing a new set of instructions (L1) that is more convenient for people to use than the set of built-in machine instructions (L0)



What is a source program?

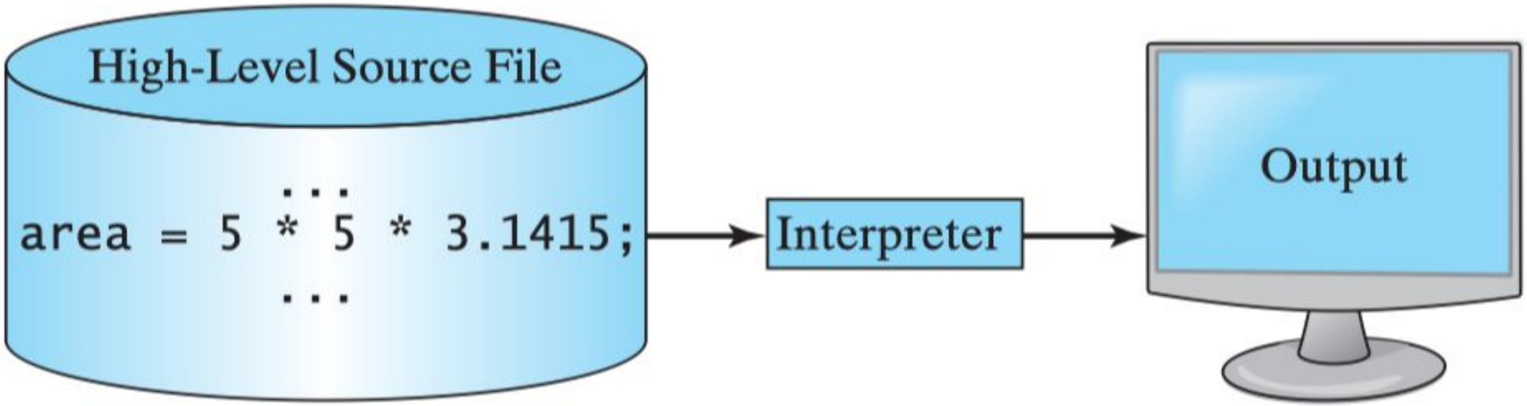
A program written in a high-level language is called a source program or source code.

Computer cannot execute a source program.
It must be **translated** into machine code for execution.

The translation can be done using another programming tool called an **interpreter** or a **compiler**.

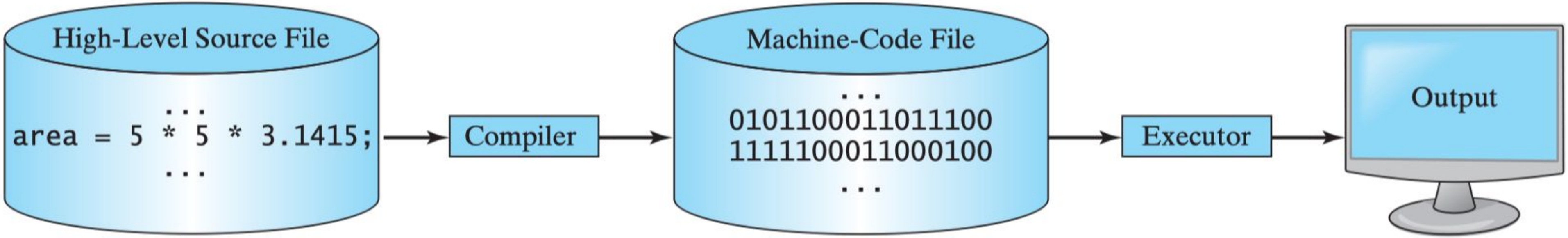
What is an interpreter?

An interpreter reads one statement from the source code, translates it to the machine code or virtual machine code at a time, and then executes it right away.

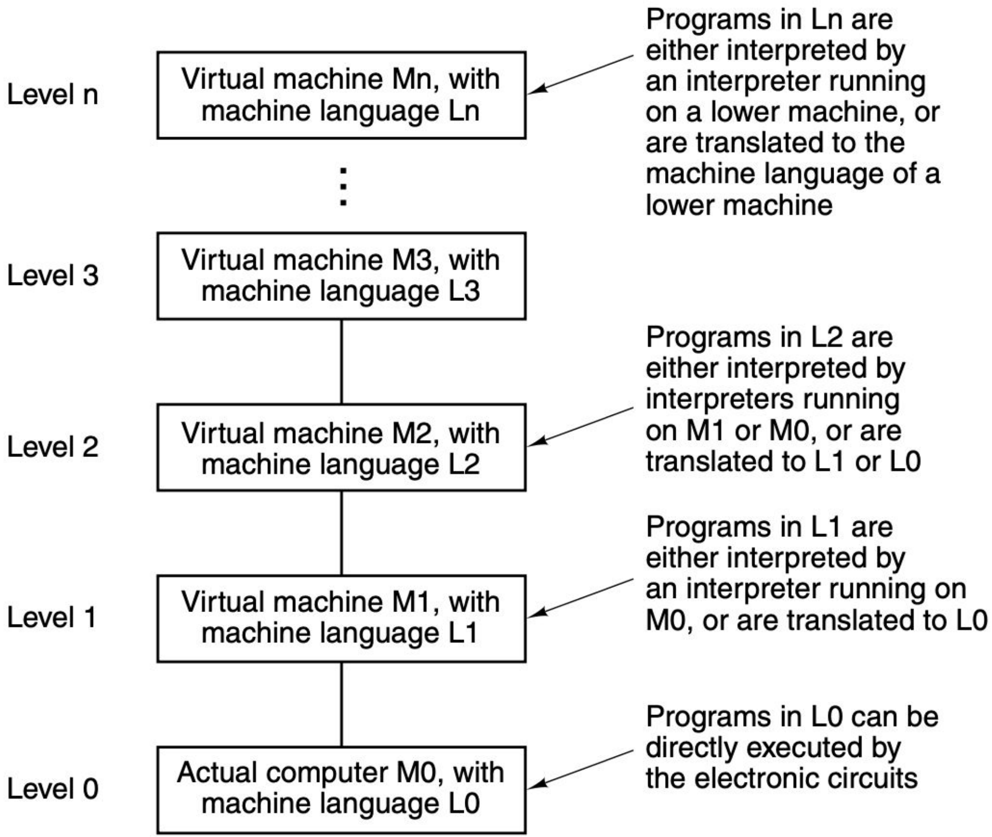


What is a compiler?

A compiler translates the entire source code into a machine-code file, and the machine-code file is then executed.



Multilevel Machine



Multilevel Machine

- Only programs written in language L0 can be **directly executed by electronic circuits**, without translation or interpretation.
- Programs written in languages L1, L2, ..., Ln must either be **interpreted or translated to a lower-level language for execution.**
- Programmers working with level n virtual machines need not be concerned with underlying interpreters or translators.
- Most programmers focus on the top level, least resembling the machine language.
- Designers of new computers or virtual machines must be familiar with all levels. Constructing machines as a series of levels is a fundamental concept in computing architecture.

Six-level Machine

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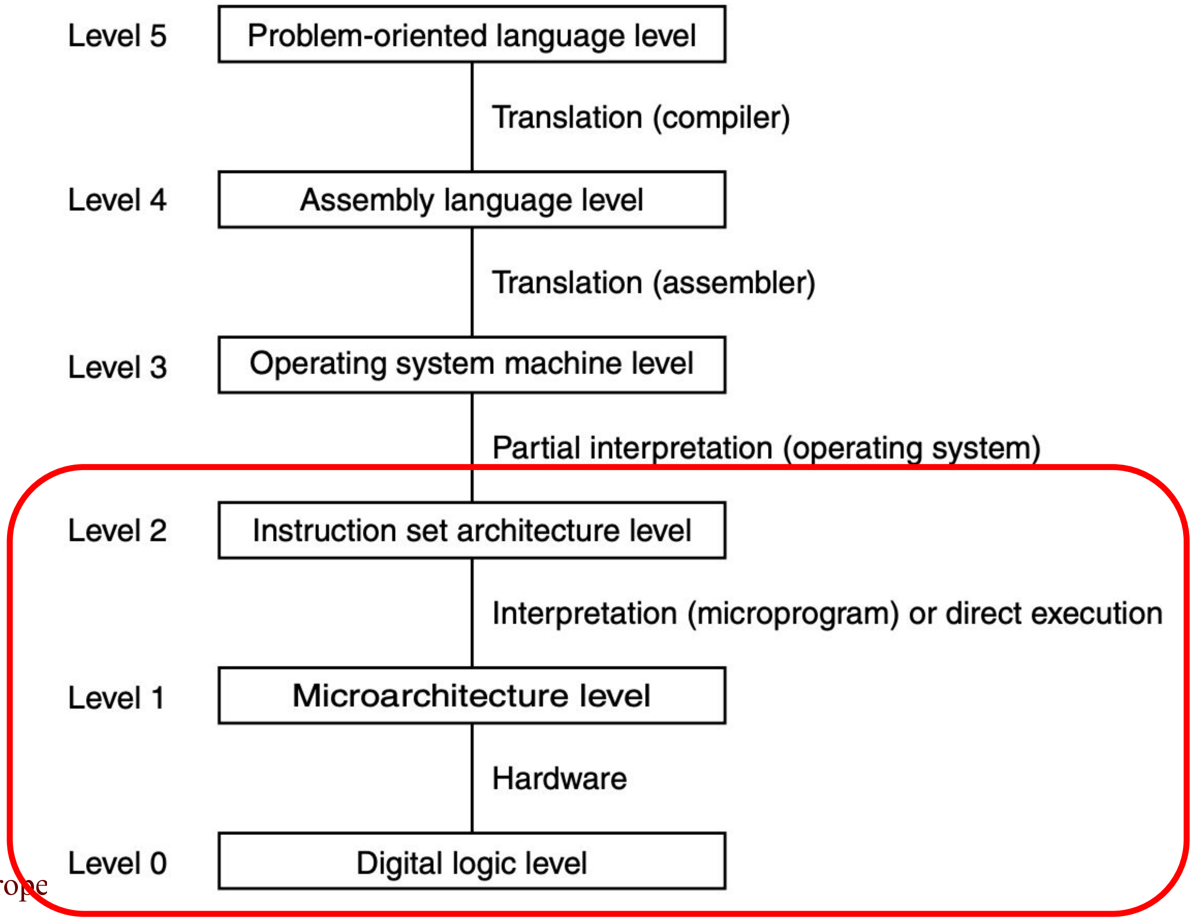

Six level Machine

- **Level 0-Digital logic level:** Gates and simple functions like AND, OR (will study)
- **Level 1- Microarchitecture level:** Local memory (registers) and ALU (Arithmetic Logic Unit)
- **Level 2- Instruction Set Architecture level:** Manual for the computer
- **Level 3- Operating system machine level:** Interacts directly with the hardware, memory, CPU, and peripherals, and providing services and interfaces for users.

Six level Machine

- **Level 4- Assembly language level:** This level provides a method for people to write programs for levels 1, 2, and 3 in a form that is not as unpleasant as the virtual machine languages themselves.
- **Level 5-Problem oriented language level:** High-level languages such as JAVA, C, C++

What is a High-Level Language?

- It is a new generation of programming languages.
- They are platform independent.
- It looks like English.

  Ex:

  area = 5 * 5 * 3.14159;

What is Assembly Language?

Assembly language uses a short descriptive word, known as a mnemonic, to represent each of the machine-language instructions.

Ex:

add 2, 3, result


Thank You!